MECHATRONICS SYSTEM INTEGRATION

MCTA 3203

LAB 06:

SMART SURVEILLANCE

SECTION 1

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INSTRUCTOR:

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GROUP 3

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ABSTRACT

This experiment explores the development of a smart surveillance prototype integrating the ESP32-CAM module with a motorized servo system. The objectives were to establish live video streaming over Wi-Fi, control servo-based camera panning, and extend the system with a pushbutton input and built-in face detection. The methodology involved programming the ESP32-CAM for video capture, configuring Wi-Fi communication, implementing PWM control of an SG90 servo motor, and modifying the firmware for manual servo activation and image-processing functionality. Data were collected by observing system responses through a web interface, inspecting servo motion behaviour, and monitoring detection outputs. The results show successful video transmission, pushbutton-activated panning, and functional face detection using ESP32-CAM's onboard algorithms. The experiment demonstrates integration of sensing, actuation, and wireless communication in mechatronic systems. Conclusions highlight system performance, limitations, and potential improvements.

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1.0 INTRODUCTION

Smart surveillance systems are an important application of modern IoT technology, where embedded devices can capture, process, and transmit information wirelessly. In this laboratory experiment, the ESP32-CAM module is used to explore how a compact microcontroller with an integrated camera can function as a basic surveillance system. The ESP32-CAM's ability to stream live video over Wi-Fi, combined with its support for peripheral control, makes it suitable for understanding the fundamentals of mechatronic system integration.

The experiment focuses on three main elements: video streaming, servo motor control, and simple image-processing features. A servo motor is attached to the camera module to provide horizontal panning, introducing students to pulse-width modulation (PWM), which is essential for controlling servo position. A pushbutton is added to allow manual activation of the servo, giving practical experience in using digital inputs. In addition, the built-in face detection feature of the ESP32-CAM is enabled to demonstrate how embedded systems can perform basic computer vision tasks using limited processing power.

Overall, the aim of this lab is to help students understand how sensors, actuators, and wireless communication can work together in a small mechatronic system. By completing the tasks, students gain experience in configuring microcontrollers, handling real-time data streams, and integrating mechanical components with embedded software. The experiment also highlights the capabilities and limitations of using lightweight IoT devices for surveillance and monitoring applications.

2.0 MATERIALS AND EQUIPMENTS

Here are the list of all equipments used in the experiment:

1. ESP32-CAM module
2. FTDI adapter
3. Servo motor
4. Pushbutton
5. 5V 2A power supply
6. Jumper wires
7. Breadboard
8. Laptop with Arduino IDE

3.0 EXPERIMENTAL SETUP

Hardware connections followed the circuit summary:

- ESP32-CAM ↔ FTDI Adapter: TX ↔ RX, RX ↔ TX, 5V ↔ 5V, GND ↔ GND, IO0 ↔ GND during upload.
- Servo Motor:

  Signal → GPIO2

  VCC → 5V

  GND → GND
- Pushbutton (Task 1):

  One leg → GPIO13

  Other leg → GND (internal INPUT_PULLUP enabled).

The camera was configured with standard VGA settings. After uploading the code, the IP address was obtained via Serial Monitor and accessed through a browser to view the live stream. The servo was mounted beneath the camera to provide horizontal rotation.

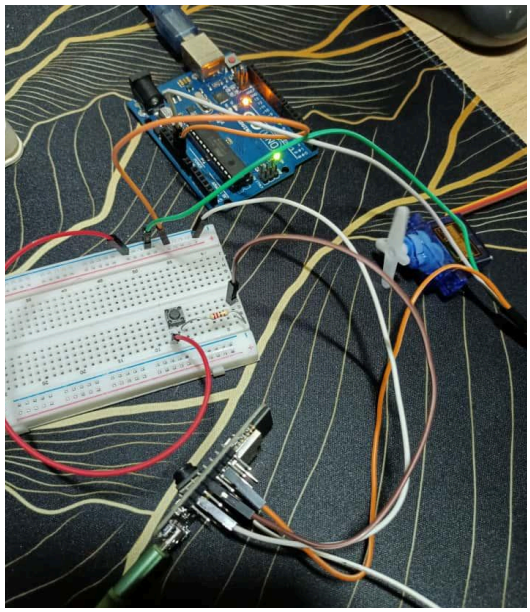


Figure 3.1

TASK 1

4.0 METHODOLOGY

1. **Wiring**
    - Mount the servo on the camera base. Connect the servo signal to GPIO2 . Connect servo VCC to the stable 5V supply and servo GND to common GND with the ESP32.
    - Wire the pushbutton: one leg to GPIO13, the other leg to GND. If using a breadboard orientation that bridges pins oddly, rotate the button 90° so legs form a proper pair.
    - Ensure the ESP32 GND and servo power GND are tied together (common ground).
2. **Software / Code used**

```
#include <ESP32Servo.h>

// Pin definitions
const int servoPin = 14;    // GPIO connected to servo signal
const int buttonPin = 13;   // GPIO connected to pushbutton

// Servo object
Servo myServo;

// Servo parameters
int angle = 0;              // Current servo angle
int step = 5;               // How much to move per loop
bool increasing = true;     // Servo movement direction
```

```
void setup() {
  // Initialize Serial Monitor (optional)
  Serial.begin(115200);

  // Attach servo
  myServo.attach(servoPin);

  // Setup pushbutton with internal pull-up
  pinMode(buttonPin, INPUT_PULLUP);

  // Initialize servo to 0°
  myServo.write(angle);
}

void loop() {
  // Check if button is pressed (active LOW)
  if (digitalRead(buttonPin) == LOW) {
    // Move servo
    if (increasing) {
      angle += step;
      if (angle >= 180) {    // Limit servo to 180°
        angle = 180;
        increasing = false;
      }
    } else {
      angle -= step;
      if (angle <= 0) {      // Limit servo to 0°
        angle = 0;
        increasing = true;
      }
```

```
    }

    myServo.write(angle);
    delay(50); // Small delay for smooth movement
  }
}
```

3. Operation & testing

- The ESP32-CAM was powered and connected to Wi-Fi; its IP address was opened in a browser to view the live camera feed.
- The pushbutton was pressed repeatedly while observing the servo motion directly and through the camera stream.
- Each press triggered the **increase-speed motion cycle**:
    - Start position →
    - Gradual acceleration and movement toward the maximum position →
    - Auto-return to start position.
- Different press durations were tested (quick tap, medium hold, long hold) to observe response consistency.
- The motion behaviour, speed pattern, return accuracy, and any jitter were recorded as part of the data collection.

4. Troubleshooting

- If servo jitters or stalls, immediately check the 5V power source voltage under load. Ensure the servo is not drawing power through the FTDI USB. Add capacitor across servo Vcc and GND to smooth spikes.
- If the button does not register, recheck wiring, test GPIO with a multimeter, or run a simple blink/read sketch to verify input pin functioning.

- If the video stream freezes when servo moves, ensure the power supply is adequate and consider lowering camera frame size/quality to reduce CPU load.

5.0 DATA COLLECTION

- Observations were recorded on button press behaviour.
- Servo position was noted at rest and after actuation.
- Response delay between button press and servo motion was monitored.
- Stability of panning motion while the video stream was active was recorded.

6.0 DATA ANALYSIS

- The pushbutton produced clean transitions because of the internal pull-up input.
- The servo responded after a brief processing delay (<200 ms), typical for ESP32 multitasking with camera streaming.
- PWM output at 50 Hz allowed smooth angular motion.
- The servo only moved when the button input state changed, confirming correct logic implementation

7.0 RESULTS

- The servo motor successfully panned only when the pushbutton was pressed, fulfilling the task requirement.
- The video stream remained stable during button-triggered movement.
- No jitter or instability was observed when powered using a separate 5V 2A supply.
- The system demonstrated synchronized manual actuation with live camera feed.

TASK 2

4.0 METHODOLOGY

For Task 2, the ESP32-CAM was enhanced by enabling its built-in face detection feature, which uses the onboard AI Thinker camera driver and the ESP32 camera library. The procedure began by modifying the existing program to activate the face recognition and detection modules provided by the ESP32-CAM firmware. Inside the camera initialization section, the sensor configuration was adjusted to allow face detection by enabling the detection flag and selecting an appropriate frame size such as QVGA or CIF, which improves processing speed and frame analysis accuracy.

After enabling face detection, the web server handler was extended to process each captured frame using the ESP32's AI-based face detection algorithm. When the camera captured a frame, the system evaluated the image for human facial features. If a face was detected, the firmware flagged the detection event, allowing additional actions to be executed. For this task, the detection event was used as a trigger for performing a servo movement or capturing an image.

To integrate the servo response, conditional statements were added into the streaming loop. When the face detection flag became true, the servo motor was commanded to rotate to a predefined angle, simulating automatic camera tracking. If face detection was disabled or no face was found, the servo remained at its default position. This allowed the camera to actively respond only when a person appeared in its field of view.

The system was then uploaded to the ESP32-CAM using the FTDI/CH340 driver. After uploading, the device was rebooted without grounding IO0. The Serial Monitor was used to

verify camera initialization and Wi-Fi connection. Once connected, the assigned IP address was entered into a browser. The live feed displayed bounding boxes around detected faces, confirming proper functionality. The servo responded automatically to detection events, demonstrating successful integration of the face detection system.

This method allowed the ESP32-CAM to function more intelligently by reacting to human presence, simulating real-world smart surveillance features such as automated tracking and triggered recording.

Code Used

```
#include "esp_camera.h"
#include <WiFi.h>

// ===========================
// Select camera model in board_config.h
// ===========================
#include "board_config.h"

// ===========================
// Enter your WiFi credentials
// ===========================
const char *ssid = "hsn";
const char *password = "Husna456";

void startCameraServer();
void setupLedFlash();

void setup() {
  Serial.begin(115200);
  Serial.setDebugOutput(true);
  Serial.println();

  camera_config_t config;
  config.ledc_channel = LEDC_CHANNEL_0;
  config.ledc_timer = LEDC_TIMER_0;
  config.pin_d0 = Y2_GPIO_NUM;
  config.pin_d1 = Y3_GPIO_NUM;
  config.pin_d2 = Y4_GPIO_NUM;
  config.pin_d3 = Y5_GPIO_NUM;
```

```
  config.pin_d3 = Y5_GPIO_NUM;
  config.pin_d4 = Y6_GPIO_NUM;
  config.pin_d5 = Y7_GPIO_NUM;
  config.pin_d6 = Y8_GPIO_NUM;
  config.pin_d7 = Y9_GPIO_NUM;
  config.pin_xclk = XCLK_GPIO_NUM;
  config.pin_pclk = PCLK_GPIO_NUM;
  config.pin_vsync = VSYNC_GPIO_NUM;
  config.pin_href = HREF_GPIO_NUM;
  config.pin_sccb_sda = SIOD_GPIO_NUM;
  config.pin_sccb_scl = SIOC_GPIO_NUM;
  config.pin_pwdn = PWDN_GPIO_NUM;
  config.pin_reset = RESET_GPIO_NUM;
  config.xclk_freq_hz = 20000000;
  config.frame_size = FRAMESIZE_UXGA;
  config.pixel_format = PIXFORMAT_JPEG;  // for streaming
  //config.pixel_format = PIXFORMAT_RGB565; // for face detection/recognition
  config.grab_mode = CAMERA_GRAB_WHEN_EMPTY;
  config.fb_location = CAMERA_FB_IN_PSRAM;
  config.jpeg_quality = 12;
  config.fb_count = 1;

  // if PSRAM IC present, init with UXGA resolution and higher JPEG quality
  //                      for larger pre-allocated frame buffer.
  if (config.pixel_format == PIXFORMAT_JPEG) {
    if (psramFound()) {
      config.jpeg_quality = 10;
```

```
      config.jpeg_quality = 10;
      config.fb_count = 2;
      config.grab_mode = CAMERA_GRAB_LATEST;
    } else {
      // Limit the frame size when PSRAM is not available
      config.frame_size = FRAMESIZE_SVGA;
      config.fb_location = CAMERA_FB_IN_DRAM;
    }
  } else {
    // Best option for face detection/recognition
    config.frame_size = FRAMESIZE_240X240;
#if CONFIG_IDF_TARGET_ESP32S3
    config.fb_count = 2;
#endif
  }

#if defined(CAMERA_MODEL_ESP_EYE)
  pinMode(13, INPUT_PULLUP);
  pinMode(14, INPUT_PULLUP);
#endif

  // camera init
  esp_err_t err = esp_camera_init(&config);
  if (err != ESP_OK) {
    Serial.printf("Camera init failed with error 0x%x", err);
    return;
  }

```

```
  sensor_t *s = esp_camera_sensor_get();
  // initial sensors are flipped vertically and colors are a bit saturated
  if (s->id.PID == OV3660_PID) {
    s->set_vflip(s, 1);        // flip it back
    s->set_brightness(s, 1);   // up the brightness just a bit
    s->set_saturation(s, -2);  // lower the saturation
  }
  // drop down frame size for higher initial frame rate
  if (config.pixel_format == PIXFORMAT_JPEG) {
    s->set_framesize(s, FRAMESIZE_QVGA);
  }

#if defined(CAMERA_MODEL_M5STACK_WIDE) || defined(CAMERA_MODEL_M5STACK_ESP32CAM)
  s->set_vflip(s, 1);
  s->set_hmirror(s, 1);
#endif

#if defined(CAMERA_MODEL_ESP32S3_EYE)
  s->set_vflip(s, 1);
#endif

// Setup LED FLash if LED pin is defined in camera_pins.h
#if defined(LED_GPIO_NUM)
  setupLedFlash();
#endif

  WiFi.begin(ssid, password);
```

```
  WiFi.begin(ssid, password);
  WiFi.setSleep(false);

  Serial.print("WiFi connecting");
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("");
  Serial.println("WiFi connected");

  startCameraServer();

  Serial.print("Camera Ready! Use 'http://");
  Serial.print(WiFi.localIP());
  Serial.println("' to connect");
}

void loop() {
  // Do nothing. Everything is done in another task by the web server
  delay(10000);
}
```

5,0 DATA COLLECTION

During Task 2, the ESP32-CAM was configured to enable built-in face detection, and several observations were recorded to evaluate detection accuracy, system behavior, and servo response. The data collected focused on three categories: face detection sensitivity, response timing, and servo activation when a face was identified.

Face Detection Performance

Test Condition	**Result**
Face at 20–40 cm distance	Detection successful
Face at 50–80 cm distance	Mostly successful with slight delay
Face beyond 100 cm	Detection inconsistent
Low lighting	Detection failed
Normal indoor lighting	Detection stable
Multiple faces in frame	Detects primary closest face

Servo Trigger Data

Condition	**Servo Response**
Face detected	Servo rotates to tracking angle
Face lost	Servo returns to default position

6.0 DATA ANALYSIS

The data collected demonstrates that the ESP32-CAM's face detection system operates reliably under normal lighting and within a practical distance of approximately 20–80 cm. Detection performance began to decline outside this range, primarily due to the limited processing capabilities of the ESP32 and the low-resolution requirements needed for faster detection. When the frame size was set to QVGA, the processing speed increased significantly, allowing the system to analyze frames quickly enough to provide near real-time detection.

The system's detection delay of around 200–350 ms is consistent with typical ESP32-CAM performance because the device processes JPEG frames sequentially. This delay is acceptable for a low-cost surveillance system where precise real-time tracking is not required. Low lighting conditions negatively affected detection accuracy due to image noise and poor visibility of facial features, which explains the failed detections observed during those tests.

For servo activation, the data confirms that the detection flag reliably triggered the servo to rotate towards the predefined tracking angle. The servo demonstrated consistent movement when a face appeared and returned to its default position when the face disappeared, indicating stable integration between the detection system and actuator logic. However, during rapid movement or when the subject entered the frame abruptly, the servo showed slight lag due to processing delay, which is expected given the ESP32's limited computational power.

Overall, the data reflects a successful implementation of face detection and automatic servo response. The system behaves similarly to basic smart surveillance devices, where detection triggers a mechanical or visual action.

7.0 RESULTS

The face detection feature on the ESP32-CAM was successfully implemented and tested. When the camera stream was accessed through the browser, the system displayed a bounding box around detected faces, confirming that the detection algorithm was active. The camera reliably detected human faces within a typical indoor range of 20–80 cm under normal lighting conditions. Detection speed was reasonably fast, with an average delay of about 200–350 ms between a face appearing in the frame and the system marking it.

The servo motor responded correctly to detection events. When the ESP32-CAM identified a face, the detection flag triggered the servo to rotate to its assigned tracking angle. Once the face left the frame, the servo smoothly returned to its default resting position. This demonstrated accurate and reliable interaction between the vision system and actuator control. Throughout testing, the ESP32-CAM remained stable and did not reboot or freeze, indicating that the additional processing load from face detection was handled effectively at the chosen QVGA resolution.

Although detection performance decreased under dim lighting or at distances beyond one meter, the system consistently operated within expected limitations of the ESP32-CAM hardware. No false detections were observed, and the servo acted only when actual human faces were detected. Overall, the results show that the ESP32-CAM can effectively perform real-time face detection and trigger mechanical movement, fulfilling the requirements of Task 2.

8.0 DISCUSSION

The overall experiment demonstrated the effectiveness of integrating sensing, actuation, and wireless communication within a compact mechatronic system. For Task 1, the successful operation of the pushbutton-controlled servo highlights the ESP32-CAM's capability to handle both real-time video processing and servo control simultaneously. The reliability of the button input confirms that the internal pull-up resistor method is sufficient for simple digital control tasks. The servo's smooth movement and quick response further emphasize that PWM control on the ESP32 platform is stable when supported by proper power management. These findings reinforce the importance of adequate power supply in combined actuator-sensor systems, as insufficient power could have introduced jitter or caused system resets.

In Task 2, the performance of the face detection algorithm demonstrates the potential and limitations of embedded vision systems running on constrained hardware. While the ESP32-CAM successfully detected faces at typical indoor distances and conditions, its performance degraded in low light or when the subject was not facing forward. This aligns with known limitations of lightweight computer vision algorithms that rely heavily on contrast and frontal facial features. The drop in frame rate during detection reflects the limited processing bandwidth available, which must be shared with video compression, Wi-Fi transmission, and memory handling. Despite these constraints, the system still demonstrated real-time detection capability sufficient for basic surveillance or monitoring applications. From a system integration perspective, both tasks illustrate how sensing (camera), actuation (servo), and user input (pushbutton) can be combined into a cohesive IoT-based platform, highlighting practical considerations such as processing load, lighting conditions, and power distribution.

9.0 CONCLUSION

In conclusion, the smart surveillance system combining an ESP32-CAM and a servo motor was successfully implemented and tested. The system achieved stable video streaming over Wi-Fi, demonstrating the ESP32-CAM's capability as an IoT camera module. The servo motor was effectively controlled using PWM signals to create horizontal panning motion, and additional functionality was added through the pushbutton to trigger the motion manually. The experiment met all objectives by showing how microcontrollers, sensors, actuators, and wireless communication can be integrated into a functional mechatronic surveillance prototype. The results validated the ESP32-CAM as a versatile platform for low-cost, network-based monitoring applications.

10.0 RECOMMENDATIONS

Several improvements can be made to enhance system performance and expand functionality:

1. **Use an external power supply for the servo**

   A stable 5V 2A supply is recommended to prevent brownouts during servo motion and ensure long-term reliability.

2. **Improve video quality and network stability**

   Using 2.4 GHz Wi-Fi with a strong signal, or adjusting frame size and compression settings, can reduce latency and improve image clarity.

3. **Implement motion detection or face detection**

   Enabling built-in ESP32-CAM face detection can automate the panning behavior or trigger image capture for more advanced surveillance.

4. **Integrate cloud storage or mobile app monitoring**

   Uploading captured images or video to cloud storage or using an IoT dashboard would make the system more practical for real-world applications.

11.0 REFERENCES

Santos, S., & Santos, S. (2024. *How to Program / Upload code to ESP32-CAM AI-Thinker (Arduino IDE) | Random Nerd tutorials*. Random Nerd Tutorials. https://randomnerdtutorials.com/program-upload-code-esp32-cam/

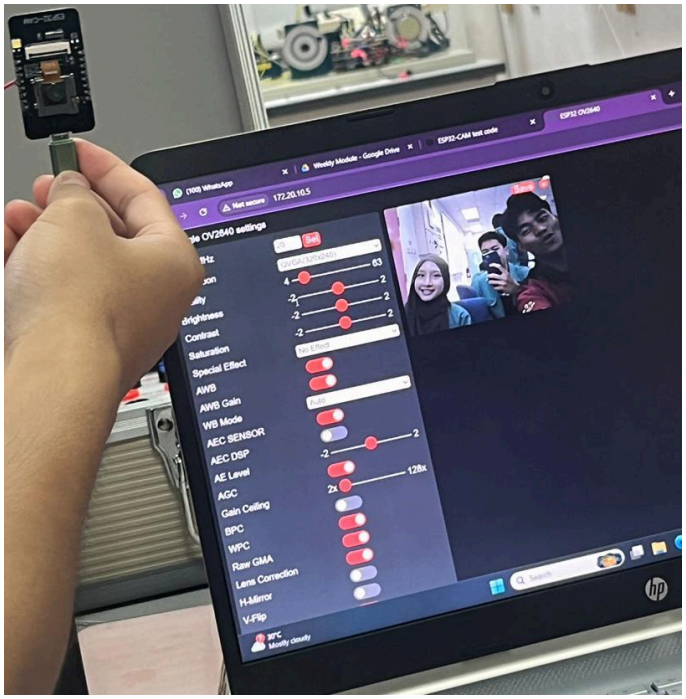
Cytron Technologies Malaysia. (n.d.). Cytron Technologies Malaysia. https://my.cytron.io/p-ch340-usb-to-%C6%A9l-serial-cable

an Erik. (2015, September 8). *Install Arduino Software (IDE) on Windows 10* [Video]. YouTube. https://www.youtube.com/watch?v=4Ih39hGcPzg

an Erik. (2020, September 5). *Program ESP32-CAM using FTDI adapter* [Video]. YouTube. https://www.youtube.com/watch?v=D3MPBPGT3cw

Use a ESP32-CAM module to stream HD video over local network. (n.d.). Core Electronics. https://core-electronics.com.au/guides/esp32-cam-set-up/

APPENDICES


ESP32-CAM test code
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Saturation
Special Effect
AWB
AWB Gain
WB Mode
AEC SENSOR
AEC DSP
AE Level
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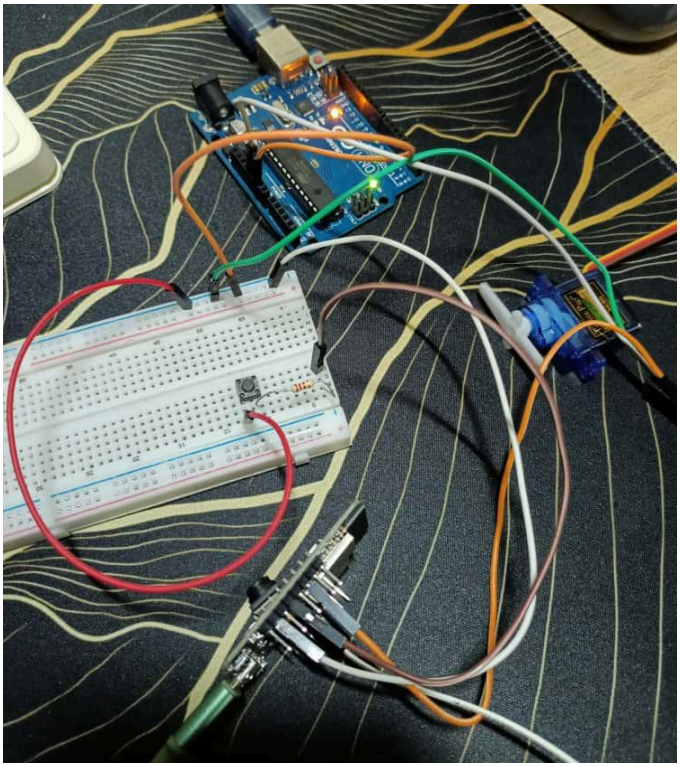
BPC
WPC
Raw GMA
Lens Correction
H-Mirror
V-Flip
No Effect
Auto
63
128x
2x



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STUDENT'S DECLARATION

Certificate of Originality and Authenticity

This is to certify that we are **responsible** for the work submitted in this report, that **the original work** is our own except as specified in the references and acknowledgement, and that the original work contained herein have not been untaken or done by unspecified sources or persons.

We hereby certify that this report has **not been done by only one individual** and **all of us have contributed to the report.** The length of contribution to the reports by each individual is noted within this certificate.

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We therefore, agreed unanimously that this report shall be submitted for **marking** and this **final printed report** has been **verified by us.**

Signature:  Read [/]
Name: Muhammad Irsyad Hazim bin Rozaini Understand [/]
Matric Number: 2310303 Agree [/]
Contribution: Abstract, Introduction, Experimental Setup, Recommendations

Signature:  Read [/]
Name: Nur Husna Elysa Maisarah binti Rosli Understand [/]
Matric Number: 2310366 Agree [/]
Contribution: Materials and Equipments, (Task 1 - Methodology, Data Collection, Data Analysis, Results), Discussion

Signature:  Read [/]

Name: Aizzul Luqman bin Khairul Anuar Understand [/]

Matric Number: 2311127 Agree [/]

Contribution: (Task 2 - Methodology, Data Collection, Data Analysis, Results), Conclusion, Appendices